TrafficSense — Multi-Agent Smart City Traffic Management System

Capstone Project | School of Computer Science & Engineering | VIT University

■ Student Information

Name	Mylapalli Abhishek Raj
Register No.	23BCE9470
Programme	B.Tech — Computer Science and Engineering
University	VIT-AP University, Amaravati
Project Title	TrafficSense: A Multi-Agent Smart City Traffic Management System
Role	Data Engineering & Integration Lead — responsible for dataset collection, preprocessing pipeline, cross-module system integration, testing, and project documentation
Academic Year	FALL SEM 2026-27
Faculty Mentor	Dr. Mohd Irfan, Assistant Professor Sr. Grade-1, SCOPE, VIT-AP University

■ Contribution Overview

My primary responsibility spanned two critical areas: (1) building the raw data collection and preprocessing pipeline that feeds the entire machine learning system, and (2) integrating all independent modules (perception, simulation, agents, dashboard) into a coherent, working system. I also owned all testing, verification, and documentation efforts for the project.

■ Detailed Technical Contributions

1. Dataset Research & Raw Data Collection

- Conducted systematic literature review to identify suitable Indian traffic datasets: IDD (Indian Driving Dataset), DriveIndia, DATS-2022, and Roboflow Universe collections.
- Evaluated each dataset for class coverage, annotation quality, image diversity, and licensing compatibility.
- Authored the data acquisition documentation and README files in data/raw/ directories.
- Managed the Roboflow API integration process and obtained the team's API credentials.

2. Data Preprocessing Pipeline

- Built src/perception/data_pipeline.py — a complete preprocessing pipeline converting raw multi-format annotations (Pascal VOC XML, COCO JSON) into YOLO TXT format.
- Implemented dataset splitting logic (70% train / 20% val / 10% test) with stratified sampling to ensure class balance.
- Wrote the dataset verification script (tests/verify_datasets.py) to check for corrupt images, missing labels, and class distribution statistics.
- Curated the final unified Indian dataset (data/processed/unified_indian/) — the standardised dataset used for all training.

3. System Integration

- Designed and implemented the end-to-end integration between the perception layer (YOLO) and the multi-agent control layer.
- Built src/perception/convert_real_states.py — converts raw YOLO detection outputs into structured CityFlow-compatible state vectors for the agent system.
- Resolved all cross-module import path issues (sys.path management, PROJECT_ROOT resolution across all dashboard pages).
- Implemented the mock Ollama server (tests/mock_ollama.py) for testing agent LLM calls without requiring a live LLM service.
- Ensured all modules share a single unified configuration (src/config.py) — no hardcoded paths.

4. Testing & Quality Assurance

- Wrote and maintained the full test suite under tests/ covering dataset validation, model loading, agent state transitions, and simulation step execution.
- Performed end-to-end integration tests: uploaded test videos to the dashboard and verified detection accuracy across all three model modes.
- Identified and documented 12+ bugs across the codebase during the audit phase, including encoding errors, path resolution bugs, and Streamlit API deprecations.
- Validated that the CityFlow simulation completes 360 steps in under 20 minutes.

5. Documentation & Deployment

- Authored the project README.md with complete setup, installation, and usage instructions.
- Managed all Git commits, branch strategy, and GitHub repository maintenance throughout the project.
- Wrote requirements.txt and ensured reproducible environment setup.
- Maintained the implementation plan and tracked progress against milestones.

■ Deliverables & Files

File / Module	Description	Status
src/perception/data_pipeline.py	Raw data preprocessing + format conversion	✔ Complete
src/perception/convert_real_states.py	Detection → agent state converter	✔ Complete
tests/verify_datasets.py	Dataset health check script	✔ Complete
tests/mock_ollama.py	Mock LLM server for offline testing	✔ Complete
tests/test_integration.py	End-to-end integration test suite	✔ Complete
data/processed/unified_indian/	Curated, split, verified training dataset	✔ Complete
README.md	Full project setup and usage documentation	✔ Complete
requirements.txt	Reproducible Python dependency list	✔ Complete

■ Technologies & Tools Used

Language	Python 3.12
Data Formats	YOLO TXT, Pascal VOC XML, COCO JSON
Testing	Python unittest + custom integration scripts
Documentation	Markdown (README, inline docstrings)
Version Control	Git + GitHub (branch management, PR reviews)
Dataset Platform	Roboflow Universe, Kaggle, IDD Portal
Environment	Windows 11 + WSL2 Ubuntu 22.04

■ Declaration

I, Mylapalli Abhishek Raj (23BCE9470), hereby declare that the contributions described in this report — including raw data collection, preprocessing pipeline, system integration, testing, and documentation — are my individual work carried out as part of the TrafficSense Capstone Project.

Signature: _____ Date: _____